

6P C01 2022

E1

$$V_{GS1} = V_{DD} - I_{D1} R_{S1} \rightarrow I_{D1} = \frac{V_{DD} - V_{GS1}}{R_{S1}} = I_{DSS} \left(1 - \frac{V_{GS1}}{V_{th1}}\right)^2$$

$$V_{GS1} = -0.52 \text{ V}$$

$$V_{GS1} = 1.333 \text{ V} > V_{th1} = 0.5 \text{ V} \quad \text{OK}$$

$$I_{D1} = I_{DSS} \left(\frac{1 - V_{GS1}}{V_{th1}}\right)^2 = 1.653 \text{ mA}$$

$$V_{GS2} = I_{D1} R_{S1} - V_{DD} = -1.363 \text{ V}$$

$$I_{D2} = I_{DSS} \left(1 - \frac{V_{GS2}}{V_{th2}}\right)^2 = 1.787 \text{ mA}$$

$$V_{DS1} = V_{DD} - I_{D1} R_{S1} = -V_{GS2} = 1.363 \text{ V}$$

$$V_{DS2} = -(V_{DD} - I_{D2} R_{D2}) = I_{D2} R_{D2} - V_{DD} = -2.32 \text{ V}$$

$$1) M1(I_{D1}, V_{DS1}, V_{GS1}) = (1.653 \text{ mA}, 1.363 \text{ V}, 1.333 \text{ V})$$

$$M2(I_{D2}, V_{DS2}, V_{GS2}) = (1.787 \text{ mA}, -2.32 \text{ V}, -1.363 \text{ V})$$

$$2) P = V_{DD} (I_{AP} + I_{D1} + I_{D2}) = 17.2 \text{ mW}$$

negligible
≈ 0

CS and CF shorted

small signal analysis

feedback topology: current mixing, voltage sensing (shunt)

Norton equivalent

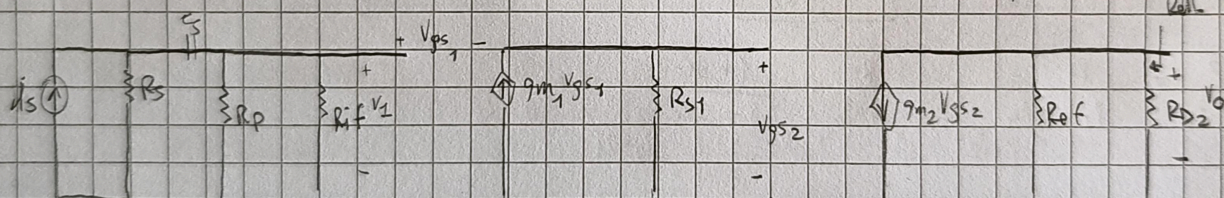
$$I_s = \frac{V_s}{R_s}$$

output network

$$R_{if} = R_F$$

$$R_{of} = R_F$$

$$\beta = -\frac{1}{R_F}$$



$$A_d = \frac{V_o}{V_s} \Big|_{\beta=0} = \alpha_i \cdot \alpha_{CD} \cdot \alpha_{CS} = -45.729 \text{ K}\Omega$$

$$g_{m1} = \frac{2I_{D1}}{V_{GS1} - V_{th1}} = 3.37 \cdot 10^{-3} \text{ S}$$

$$\alpha_{CS} = \frac{V_o}{V_{GS2}} \Big|_{\beta=0} = -g_{m2} (R_{of} \parallel R_{D2}) = -6.12 \text{ [V/V]}$$

$$g_{m2} = \frac{2I_{D2}}{V_{GS2} - V_{th2}} = 4.41 \cdot 10^{-3} \text{ S}$$

$$\alpha_{CD} = \frac{V_{GS2}}{V_{S1}} \Big|_{\beta=0} = \frac{g_{m1} R_{S1}}{1 + g_{m1} R_{S1}} = 0.897 \text{ [V/V]}$$

$$A_{d1} = -45.729 \text{ K}\Omega$$

$$\alpha_i = \frac{V_{S1}}{V_s} \Big|_{\beta=0} = R_s \parallel R_p \parallel R_{if} = 8.33 \text{ [K}\Omega\text{]}$$

$$A_F = \frac{A_d}{1 + \beta A_d} = -31.379 \text{ [K}\Omega\text{]} \rightarrow A_V = \frac{V_o}{V_s} = A_F \cdot \frac{1}{R_s} = -3.138 \text{ [V/V]}$$

$$R_{in}^a = \alpha_i = 8.33 \text{ [K}\Omega\text{]}$$

$$R_{out}^a = R_{of} \parallel R_{D2} = 1.478 \text{ K}\Omega$$

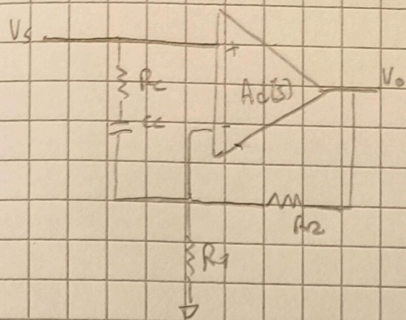
$$R_{in}^F = \frac{R_{in}^a}{1 + \beta A_d} = 5.946 \text{ K}\Omega$$

$$R_{out}^F = \frac{R_{out}^a}{1 + \beta A_d} = 1.014 \text{ K}\Omega$$

$$R_{in} = \frac{1}{\frac{1}{R_{in}^F} - \frac{1}{R_s}} = 13.343 \text{ K}\Omega$$

$$R_{out} = R_{out}^F = 1.014 \text{ K}\Omega$$

EX 2



$$A(s) = \frac{A_{d0}}{(1 + \frac{s}{\omega_1})(1 + \frac{s}{\omega_2})}$$

$$\begin{aligned} R_1 &= 33 \text{ k}\Omega \\ R_2 &= 120 \text{ k}\Omega \\ R_c &= 150 \Omega \\ C &= 330 \text{ nF} \end{aligned}$$

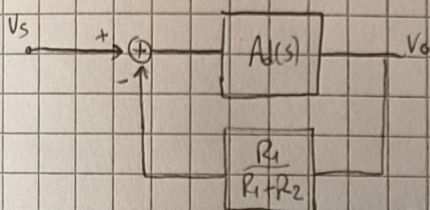
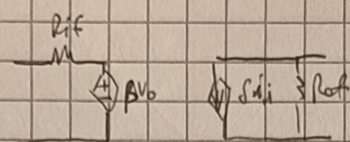
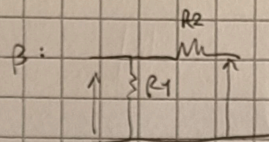
$$\begin{aligned} A_{d0} &= 10^5 \\ \omega_1 &= 2 \cdot 10^3 \text{ rad/s} \\ \omega_2 &= 6 \cdot 10^5 \text{ rad/s} \end{aligned}$$

C_c needed:

$$1) \text{ Ideal } A_v = \frac{V_o}{V_s} = 1 + \frac{Z_2}{Z_1} = 1 + \frac{R_2}{R_1} = 4.64 \text{ [V/V]}$$

2) feedback: voltage mixing, voltage sampling

$$\beta = \frac{R_1}{R_1 + R_2}$$



$$\frac{A_d(s)}{1 + A_d(s) \beta} \xrightarrow{A_d \rightarrow \infty} \frac{R_1 + R_2}{R_1} = 1 + \frac{R_2}{R_1} \quad \text{OK}$$

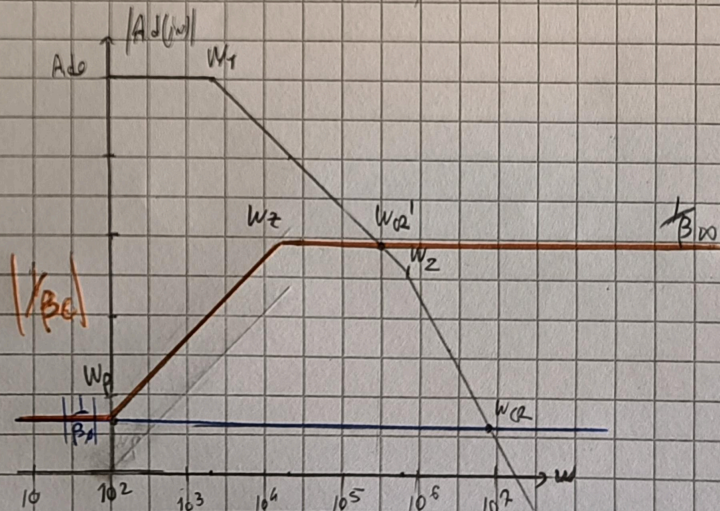
$$\beta_{\text{dB}} = \frac{R_1}{R_1 + R_2} \rightarrow \frac{1}{\beta_{\text{dB}}} = 1 + \frac{R_2}{R_1} = 4.64 = 13.33 \text{ dB}$$

low gain

$$\begin{cases} A_{d0} \omega_1 = A \times \omega_2 \\ A \times \omega_2^2 = \frac{1}{\beta} \omega_2^2 \end{cases}$$

$$\omega_{CR} = \sqrt{\frac{1}{\beta} \omega_1 \omega_2} = 5.087 \cdot 10^6 \text{ rad/s}$$

$$\varphi = 180^\circ - \arctan\left(\frac{\omega_{CR}}{\omega_1}\right) - \arctan\left(\frac{\omega_{CR}}{\omega_2}\right) = 6.75^\circ \text{ low phase margin}$$



with $R_c - C_c$: noise gain compensation

$$\beta_c = \beta_{c0} \cdot \frac{1 + s/\omega_c}{1 + s/\omega_p} = \beta_{c0} \cdot \frac{1 + s R_c C_c}{1 + s C_c R_c} \rightarrow \omega_c = \frac{1}{C_c R_c} = 2.02 \cdot 10^4 \text{ rad/s}$$

$$\omega_p = \frac{1}{C_c (R_c + R_1 \parallel R_2)} = 116.6 \text{ rad/s}$$

$$\beta_{c0} = \beta_{c0} \cdot \frac{R_c}{R_c + R_1 \parallel R_2}$$

$$\beta_{c0} = \beta_{c0} \cdot \frac{R_c}{R_c + R_1 \parallel R_2} = \frac{R_1}{R_1 + R_2} \cdot \left(\frac{R_c}{R_c + R_1 \parallel R_2} \right) = 1.24 \cdot 10^{-3}$$

$$\frac{1}{\beta_{c0}} = 805 = 58 \text{ dB}$$

$$\omega_{CR} : A_{d0} \omega_1 = \frac{1}{\beta_{c0}} \omega_{CR} \rightarrow \omega_{CR} = \beta_{c0} A_{d0} \omega_1 = 2.484 \cdot 10^5 \text{ rad/s}$$

$$\text{PM} = 180^\circ - \arctan\left(\frac{\omega_{CR}}{\omega_1}\right) - \arctan\left(\frac{\omega_{CR}}{\omega_2}\right) - \arctan\left(\frac{\omega_{CR}}{\omega_p}\right) + \arctan\left(\frac{\omega_{CR}}{\omega_c}\right) = 62.35^\circ \quad \text{OK}$$

1 dB